Venkata Lakshmi Parimala Pasupuleti

Applied AI Engineer · Python / LangGraph / RAG Pipelines · Agentic Systems · LLMOps & Eval Infrastructure · Responsible AI

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SUMMARY

Applied AI Engineer with 9 years of production software ownership and hands-on delivery of agentic AI systems from architecture to production. At Magizhs Tech Solution, designed and delivered a production LangGraph agentic AI system for an external enterprise client -- full technical ownership: LLM selection, retrieval-augmented generation (RAG) pipeline over a regulatory corpus, prompt engineering, output validation guardrails, and integration with enterprise data sources. Built AgentProbe -- an open-source LLMOps and eval platform: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, AI observability (agent tracing, metrics, multi-provider routing), and Responsible AI guardrails including red-teaming and PII handling. Built Leasel -- an 8-agent LangGraph system with RAG retrieval, structured data extraction, HITL guardrails on regulated decisions, and conditional routing: 93.6% precision, 87.9% F1 across 1,240 test cases. Seven years owning data pipelines, containerized microservices, and enterprise compliance systems at Entrata (Python, JavaScript/TypeScript, PostgreSQL, RabbitMQ, AWS, 100K+ MAU). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & Agentic Systems: Python, LangGraph, LangChain, Agentic AI, Agentic Workflows, Multi-Agent Orchestration, Large Language Models (LLMs), Generative AI, Prompt Engineering, ReAct, Tool Calling, MCP (Model Context Protocol), RAG Pipelines, Vector Databases, ChromaDB, Embeddings, HITL, Responsible AI, Guardrails, Output Validation, Hallucination Detection, LLM Evaluation, LLM-as-Judge, Agent Tracing, AI Observability, LLMOps, Fine-tuning (HuggingFace, PyTorch), NLP, BERT, spaCy

Full-Stack & Data Pipelines: Python, JavaScript, TypeScript, Java, SQL, React.js, Angular, FastAPI, Flask, REST APIs, Microservices, Data Pipelines, ETL, Data Modeling, RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, AWS (S3, EC2, RDS, Glue, Lambda), Azure, GCP, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), Git, Linux, Agile/Scrum

Enterprise & Compliance: HIPAA Compliance, PHI Data Handling, RBAC, Libsodium, Distributed Systems, Observability, Performance Engineering, Code Review, Technical Mentorship

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI Engineer (Customer-Facing Delivery)

- Designed and delivered a production LangGraph agentic AI system for an external enterprise client -- full technical ownership from architecture through deployment: LLM selection, RAG pipeline over a regulatory corpus, prompt engineering strategy, output validation guardrails, and integration with enterprise data sources.
- Built Responsible AI validation guardrails enforcing compliance rules deterministically -- non-compliant outputs blocked before delivery with grounded cited reasons; shipped an eval harness alongside the product so every output was validated before reaching the client's production environment.
- Translated customer business requirements into a working agentic solution, iterating on prompt flows and automated evaluations in an agile cadence.

Entrata (PropTech SaaS · 100K+ MAU · enterprise data pipelines · HIPAA/PHI)
Sep 2018 - Aug 2024

Senior Software Engineer

- Owned compliance-critical data pipelines for 7 years (180,000 HUD records powering housing eligibility, rent, and tax-credit calculations for 100K+ units) -- zero violations; validation, reconciliation, and rollback built into every release cycle.
- Shipped 20+ full-stack production modules (Python, JavaScript/TypeScript, React, Angular, PostgreSQL, RabbitMQ) across 120+ Agile sprints; led Docker microservices redesign (50% scalability improvement, 30% deployment time reduction).
- Led AWS S3 document migration -- Libsodium-encrypted pipelines, 100% PHI compliance, zero security incidents; built Python/Bash automation eliminating 80% of manual operational workload.
- Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and stored procedure refactoring; mentored 5+ engineers; conducted 300+ peer code reviews.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Fine-tuned a custom classification model architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented model design and evaluation results to faculty.
- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest anomaly detection across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: LLMOps Platform & AI Observability Infrastructure
2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · LangChain · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) · MCP (Model Context Protocol)

- Open-source LLMOps and eval platform with full AI observability: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, pass@k harness, agent tracing (logs/metrics/traces), multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), and Responsible AI guardrails (filters, red-teaming, rate limiting controls, PII handling). Instruments agentic workflows for reliability and quality -- the production-grade infrastructure for knowing whether an AI system is ready before it touches real users.

Leasel: 8-Agent Agentic System with RAG, HITL & Eval Infrastructure

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit · Responsible AI

- Production agentic AI system: 8 LangGraph agents with RAG retrieval, structured data extraction, prompt engineering, Responsible AI guardrails (HITL escalation on regulated decisions, safety boundary enforcement), and conditional routing. 93.6% precision, 87.9% F1 across 1,240 test cases. Architected end-to-end RAG pipeline: OCR ingestion, section-aware chunking, embeddings, indexing, and retrieval -- from sketch to deployed and observable in production.

AI on FHIR: HIPAA-Compliant Healthcare AI System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR APIs · spaCy · BERT · Docker · HIPAA · RBAC

- HIPAA-compliant AI system over structured PHI -- RBAC, access controls, PII handling, NLP-to-FHIR API translation (EN/ES), LLM + BERT + embeddings integration; reduced clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017